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Chuan Li, Ruoxi Zhong, Hao Chen & Yunwei Liu

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Inhibition or Promotion: A Study on the Impact of Relocation on the Mental Health of Left-behind Women

Chuan Li¹□, Ruoxi Zhong²□, Hao Chen¹, Yunwei Liu^{1*}

¹College of Resources and Environment, Xichang University, Xichang 615000, China.

²College of Economics, Sichuan Agricultural University, Chengdu 611130, China.

*Corresponding author(s). E-mail(s): xcc03800029@xcc.edu.cn

Contributing authors: lic@xcc.edu.cn; zhongruoxi@stu.sicau.edu.cn; chenhao@xcc.edu.cn

□These authors contributed equally to this work.

Abstract

Background China's poverty alleviation relocation program aims to improve living standards in impoverished areas through population resettlement. However, the mechanisms through which it affects the psychological well-being of left-behind women have not been clearly elucidated. Existing research predominantly focuses on the material outcomes of the policy, lacking systematic investigation into the often overlooked dimension of psychological stress.

Methods Based on field survey data collected from July to November 2024, this study employs an Ordered Probit model for baseline regression and utilizes an instrumental variable approach to address endogeneity concerns. Robustness checks are conducted through replacing dependent variables and excluding specific samples. Mediation effect models analyze the intermediary roles of consumption costs and livelihood risks, while a moderation effect model examines the buffering effect of support policies.

Results The study finds that relocation significantly increases work stress ($\beta=1.342$, $p<0.01$) and life stress ($\beta=0.613$, $p<0.01$) among left-behind women. Instrumental variable estimates further confirm the robustness of these negative effects. Mechanism analysis demonstrates that relocation exacerbates psychological stress through two pathways: increasing household consumption costs and elevating perceived livelihood risks. Moderating effect analysis indicates that support policies can effectively mitigate the negative psychological impact of relocation. Heterogeneity analysis reveals that low-income households and those lacking industrial support experience greater psychological pressure.

Conclusions This study reveals the inhibitory effect of relocation on the mental health of left-behind women. While the relocation policy improves material conditions, it may negatively affect the

mental health of vulnerable groups due to livelihood transition pressures and social network disruption. The study recommends that subsequent policies should establish a tiered subsidy system, improve mental health services, implement differentiated support measures, and develop local industries along with universal childcare services to build a sustainable psychological support system.

Keywords Relocation □ Left-behind women □ mental health □ psychological stress

Introduction

Population migration and resettlement have long served as crucial policy tools for poverty alleviation, addressing ecological vulnerability, and promoting balanced regional development [1-2]. Scholarship continues to debate whether resettlement, from developmental relocation in South Asia and Latin America to post-disaster resettlement in Africa, functions as an effective poverty reduction strategy or, while eliminating previous poverty, creates new social and psychological challenges [3]. The impact of migration and resettlement on individual mental health exhibits complexity and duality. On one hand, improvements in the living environment, enhanced access to public resources, and expanded

social opportunities may significantly boost subjective well-being and psychological security [4-6]. On the other hand, the disruption of social networks, stress from community cultural adaptation, and the reconfiguration of family roles can trigger psychological issues such as anxiety, depression, and loneliness [7-8]. This coexisting nature of “inhibition” and “promotion” indicates that the effects of resettlement policies are not unidimensional but are deeply influenced by the interplay of individual circumstances, social relationships, and gender roles.

The gender factor is particularly critical within the context of rural Chinese society [9]. Long constrained by traditional gender divisions of labor and structural inequalities, rural women often bear the primary responsibility for family care, intergenerational support, and emotional maintenance. When male laborers migrate for work, left-behind women shoulder immense pressure in sustaining household operations, experiencing not only heavy labor burdens but also high psychological vulnerability due to a lack of social support and limited economic opportunities [10-11]. In particular, in the resettlement project for poverty alleviation implemented during China’s poverty alleviation campaign, the mental health risks of this group urgently require attention. Between 2016 and 2020, China relocated over 9.6 million poor

people through resettlement, while 5 million non-poor people were also relocated simultaneously in central and western regions, equivalent to the population size of a medium-sized country. Western China is a concentrated area for such resettlement, where the number of poor people reached 6.64 million, accounting for 67.7% of the total relocated population, and 4.23 million non-poor people were also relocated through the same program. Resettlement is mainly carried out in centralized settlements, accommodating 12.44 million people, which accounts for 76.4% of the total relocated population. Compared with the general resettled group, left-behind women face more prominent issues of livelihood transition pressure and lack of social support, and the inhibitory effect of resettlement on their mental health may be more significant. Existing research mostly focuses on material dimensions such as economic improvement and enhanced living conditions, with less systematic investigation into the impact of resettlement on migrants' mental health, and even fewer detailed analyses concerning gender differences and family role dimensions.

However, existing research predominantly focuses on material dimensions such as economic improvement and enhanced living conditions, with less systematic investigation into the mental health impacts of relocation on migrants. There is a particular scarcity of

detailed analysis concerning gender differences and family role dimensions. Focusing on left-behind women within China's relocation program, this study examines whether resettlement promotes their mental health by improving environmental and resource conditions, or inhibits it by weakening social support networks and intensifying family pressures. This investigation not only helps broaden the overall understanding of the social effects of relocation policies but also offers practical implications for formulating resettlement support policies that are more gender-sensitive and mental health-oriented. To address the above questions and clearly present the research context, this study is structured as follows: The second part consists of a literature review and research hypotheses. Based on sorting out the relationship between migration and mental health as well as the multidimensionality of poverty, the research hypotheses of this paper are proposed by integrating stress theory and the family stress model; The third part introduces the data sources, variables, and model specifications; The fourth part presents the empirical results and analysis, including baseline regression, endogeneity treatment, robustness checks, mechanism analysis, and moderation effect tests; The fifth part contains conclusions and policy implications, which thoroughly discuss the heterogeneous results

and put forward targeted policy recommendations; Finally, the research limitations are pointed out.

Literature and hypotheses

Literature review

Poverty is a complex concept with multidimensional attributes, whose connotation goes far beyond the single category of income deprivation. Elements such as asset stock, livelihood stability, and social capital collectively constitute the identification and measurement system of poverty. Income poverty is often temporary and volatile, while asset poverty can better reflect a household's long-term ability to resist risks, including tangible assets such as land, housing, and means of production, as well as intangible assets such as social networks and professional skills [12-13]. Although resettlement may improve households' material conditions in the short term through housing subsidies and infrastructure upgrades, if it fails to effectively promote the substantial accumulation of household assets or alleviate the disruption of existing social capital caused by relocation, resettled households may still fall into the predicament of "improved income but fragile wealth." This form of poverty in the wealth dimension may exert a persistent and invisible impact on the mental health of left-behind women through

channels such as livelihood uncertainty, social comparison pressure, and future anxiety. Therefore, when evaluating the impact of resettlement policies, it is necessary to go beyond the single dimension of income and incorporate multiple indicators such as wealth, social capital, and psychological belonging, so as to fully understand the complex effects of relocation on the psychological well-being of vulnerable groups [14-15].

Migration constitutes a dynamic process involving adaptation and resocialization. Existing literature generally acknowledges a complex and often contradictory relationship between migration and mental health [16-18]. On one hand, migration may enhance an individual's sense of well-being and mental health by expanding economic opportunities and improving access to public services [6, 19]. On the other hand, migration can coincide with crises related to livelihood strategy transformation, cultural adaptation, and identity, potentially triggering psychological stress [9, 20, 21]. Consequently, scholars frequently emphasize the “double-edged sword effect” of migration, where the outcome typically depends on the combined influence of resettlement policy design, support systems during the migration process, and individual factors such as gender and family roles [22-24].

Economic and social resources represent significant

dimensions influencing mental health. Substantial empirical research has explored disparities in the mental health of migrant populations [25-27]. Studies on international migration in developed countries find that immigrants often face elevated risks of mental illness due to social exclusion, employment instability, and cultural barriers [28]. In contrast, research on internal migration within developing countries indicates that while migration can improve economic status, it frequently exacerbates family separation and lack of social support, thereby undermining mental health [29-30]. Concurrently, the public health field increasingly emphasizes the intersection of migration and health, recognizing mental health not only as a neglected dimension in migration studies but also as a crucial metric for evaluating the success of relocation policies [31]. Research incorporating a gender perspective is particularly vital. Women often occupy more vulnerable positions during migration and resettlement. The process can intensify their domestic caregiving and emotional burdens, while also adding employment responsibilities aimed at alleviating family financial crises. Left-behind women are especially susceptible to mental health issues [32-35]. Furthermore, consumption inequality can trigger psychological stress and negative emotions, thereby impairing mental health. In the context

of resettlement, this vulnerability is further amplified: the transformation of livelihood patterns brought about by relocation imposes higher requirements on women's skill reserves, while the family care responsibilities entrusted by traditional gender division of labor have not been alleviated accordingly. Under the dual pressures, the mental health of left-behind women is more susceptible to impacts. Evidently, the mental health of left-behind women is subject to pressures from multiple factors [36-38].

China's relocation program has significantly improved housing conditions and economic income for participants, while also fostering social integration and intergenerational mobility [39-40]. However, the relocation process often leads to the dissolution of familiar social networks, the weakening of traditional cultural ties, and a diminished sense of community belonging, consequently causing psychological discomfort and social alienation [41-43]. In rural China, with husbands often working away from home for extended periods, left-behind women must single-handedly manage multiple responsibilities, including household chores, child-rearing, and elder care. This situation severely impacts their physical and mental well-being [44-47]. Empirical studies further indicate that the physical and mental quality of life for left-behind women is significantly lower than that of their non-left-behind counterparts

[48-49]. Nevertheless, existing research lacks an in-depth exploration of mental health, particularly a differential analysis considering gender and family roles. Few studies reveal how relocation influences the mental health status of left-behind women by altering household livelihoods, consumption patterns, and the division of family roles.

Research hypotheses

For left-behind women experiencing the major life event of relocation, the impact on mental health is not only reflected in the work domain but also deeply permeates their daily lives and family roles. Perceived stress is their most immediate and subjective psychological response when coping with a series of challenges brought about by relocation, serving as the primary signal of mental health risks [50-51]. Using stress as an indicator for assessing the mental health of left-behind women can more sensitively and accurately capture the short-to-medium term impacts of policy interventions on individual psychological well-being.

Relocation exerts a direct effect on the psychological stress of left-behind women through multidimensional environmental changes. Theoretically, positive factors such as improved living conditions and enhanced accessibility of public services have the

potential to alleviate psychological stress. However, based on stress theory, stress stems from an individual's perception of imbalance between environmental demands and their own coping capabilities [52]. Considering the actual situation of left-behind women, the disruptive impacts brought by relocation are more prominent and unavoidable: first, the transformation of livelihood patterns from self-sufficient agricultural production to market-oriented non-agricultural employment leaves left-behind women who have long relied on farming facing practical dilemmas such as devaluation of core skills and employment competition, directly intensifying occupational stress; second, the original social networks in acquaintance societies collapse due to geographical migration, and the reconstruction of social capital in new communities requires a relatively long cycle. Under the dual absence of family care and emotional support, left-behind women are prone to loneliness and anxiety; third, the rigid rise in monetized living costs further amplifies household financial pressure. Overall, the negative impacts of relocation far outweigh the positive effects of environmental improvements. Based on this, Research Hypothesis H1 is proposed:

H1: Relocation will exacerbate the psychological stress of left-behind women and exert an inhibitory effect on their mental

health.

As households transition from a semi-subsistence model to a fully monetized lifestyle, rigid increases in daily expenses such as utilities and food directly increase the financial burden on families. The family stress model suggests that economic pressure is the core transmission mechanism of family dysfunction and psychological distress among family members [53-54]. As the primary managers of household expenses, left-behind women are particularly sensitive to this shift towards a “cash-based expenditure” lifestyle. This sensitivity can lead to persistent financial anxiety, subsequently generating significant psychological stress. Therefore, consumption costs may serve as an important channel through which relocation influences psychological stress. Research Hypothesis H2 is proposed:

H2: Relocation increases the psychological stress of left-behind women by raising household consumption costs.

Risk perception and stress theory indicate that individuals' perception of future uncertainty will activate stress responses, especially when household livelihood security is threatened [55]. The transition from stable farming livelihoods to market-based non-farm employment involves multiple risks, including skills mismatch, employment instability, and income volatility.

Left-behind women's concerns about household livelihood security transform into a heightened sense of uncertainty about the future. This perception of risk becomes particularly acute when the primary household labor force is engaged in highly volatile sectors such as construction, easily leading to a state of persistent psychological tension. Research Hypothesis H3 is proposed:

H3: Relocation increases the psychological stress of left-behind women by elevating the household's perception of livelihood risk.

According to the stress-buffering model, external support can effectively mitigate the psychological impact of negative life events. The conservation of resources theory also emphasizes that external resources such as social support, economic assistance, and public services can enhance individuals' ability to cope with stress [56]. As a crucial external resource for relocated families, support policies can directly alleviate economic pressure by promoting employment or providing temporary subsidies. These policies can also weaken the negative effects of relocation by organizing community activities to facilitate social integration. Research Hypothesis H4 is proposed:

H4: Support policies play a moderating role in the relationship between relocation and psychological stress. Left-behind women who receive more policy support will experience a weaker stress

effect from relocation.

Figure 1 shows the research mode.

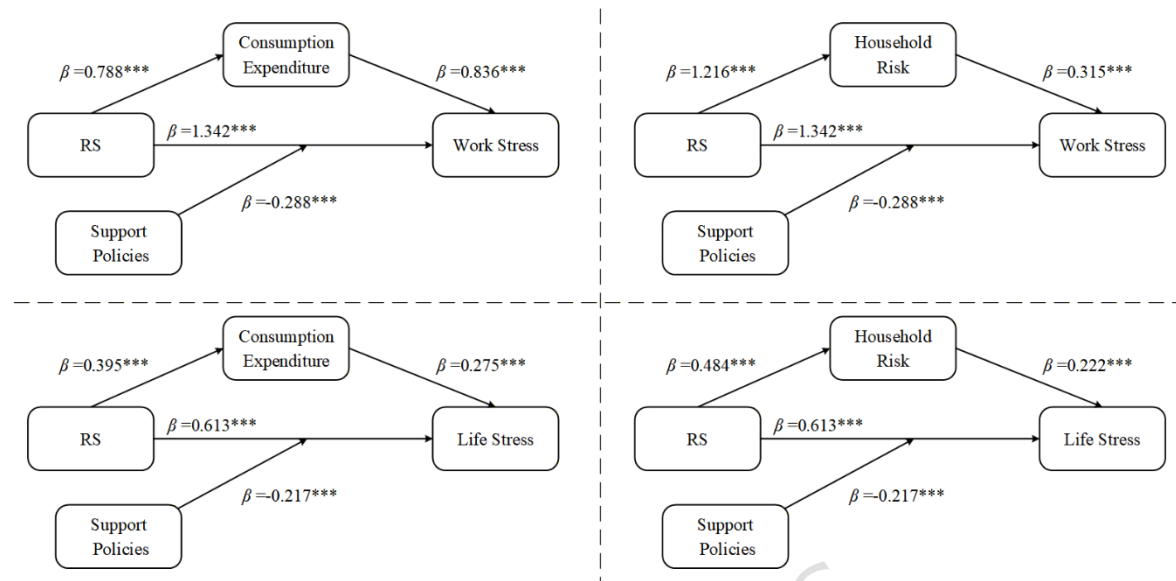


Fig.1 The research mode

Materials and methods

Data

The microdata used in this study is derived from a field survey on the livelihood transformation of resettled households and follow-up support for left-behind women, conducted by the research team between July and November 2024. The survey covered the core areas of China's contiguous poor areas and their surrounding regions, which are also among the regions with the largest scale of resettled populations and the heaviest resettlement tasks under the poverty alleviation relocation program. To reduce potential limitations and biases in the survey data, anonymous processing was adopted throughout the research, with samples

identified by community codes and household codes. Considering the respondents' cultural backgrounds and language habits, the survey was conducted through face-to-face interviews. Local college and graduate students who were familiar with the local dialect and received professional training served as investigators, interpreting each questionnaire item to the respondents one by one and filling in the questionnaire content directly based on their answers. For logically contradictory or ambiguous responses, investigators conducted on-site follow-up inquiries to confirm, minimizing information bias to the greatest extent.

The survey aimed to cover typical resettlement areas, and the sampling design followed the principles of stratified, multi-stage, and random sampling to ensure the representativeness of the samples. Specifically, stratification was carried out according to regional types and resettlement methods to ensure the representativeness of samples in each stratum. Within each stratum, left-behind women were randomly selected from resettlement communities to form the treatment group. To construct a comparable control group and enhance the ability to identify the impact of relocation, the research team also randomly selected eligible non-resettled left-behind women from natural villages with similar geographical locations, economic conditions, and socio-cultural environments as the comparison group. This design ensured that the non-resettled samples were mainly from regions similar to the treatment group in terms of living

background, policy environment, and social structure, thereby improving the comparability of the impact of relocation on the mental health of resettled and non-resettled women.

The survey distributed 800 questionnaires and retrieved 791. After rigorous data cleaning that removed samples with missing key variables or logical inconsistencies, 779 valid questionnaires remained, yielding an effective response rate of 97.4%. The final valid sample provides good representation of the study population and offers a reliable data foundation for subsequent empirical analysis.

Variables

Explained variable

From the perspective of classic psychological and sociological theories, individual psychological stress exhibits domain specificity. Mainstream stress research generally divides stress into two core dimensions: livelihood-related stress and life-related stress [57-58]. This framework is concise, stable, and applicable across groups, making it particularly suitable for vulnerable rural groups with relatively limited educational backgrounds who struggle to complete complex multi-dimensional scale surveys [59-60]. Compared with comprehensive psychological indicators such as depression, anxiety, and somatization, work stress and life stress can more directly reflect realistic stressors caused by specific life

events, and have a more direct correlation and stronger explanatory power with external shocks such as relocation. After relocation, left-behind women face the dual role conflict of being “the main bearers of household livelihood” and “the core caregivers of the family”, and their stress sources are highly concentrated in two areas: first, work stress related to livelihood and employment, including unstable employment, skill mismatch, heavy labor burden, and income uncertainty; second, life stress related to family life, including rising expenses, increased caregiving burden, family separation, and difficulties in adapting to the living environment [61]. Therefore, selecting work stress and life stress can maximize the capture of the main contradictions and core characteristics of psychological stress among this group.

From the perspective of the practical constraints of the survey respondents, the sample of this study consists of left-behind women resettled through poverty alleviation relocation in China’s extremely poor areas. This group generally has low educational levels, with most respondents being illiterate or semi-illiterate, and daily communication mainly in local dialects. The survey relies on investigators proficient in both dialect and Mandarin to conduct face-to-face interviews, requiring respondents to accurately distinguish gradient concepts such as “low stress”, “moderate stress”, and “high stress”. Work stress stems from the transformation of livelihood patterns from agriculture to non-agricultural employment. Left-behind women face a series of

continuously changing challenges such as skill learning, employment competition, and income volatility, and the intensity of stress shows obvious gradient differences. Therefore, the use of an ordinal scale can accurately capture changes in their stress levels.

In contrast, life stress arises from the transformation of household daily life patterns after relocation. Before relocation, rural households were mainly self-sufficient, with extremely low proportions of rigid expenses such as water, electricity, and gas, resulting in low daily living costs. After relocation, households have fully transitioned to a monetized lifestyle, with a sharp increase in various rigid expenses. As the main managers of household expenses, left-behind women are confronted with sudden and holistic stress shocks. The core of this shock lies in the essential change of “whether stress exists” rather than the gradient difference of “how high the stress level is”. Therefore, the use of a binary variable can accurately match this core characteristic of life stress.

From the perspective of the core research objectives, this study focuses on revealing the impact of relocation on the mental health of left-behind women, with particular attention to whether relocation triggers the occurrence of life stress and enhances work stress. The binary measurement method can directly reflect the “trigger relationship” between relocation and life stress, complementing the intensity measurement of work stress. From the two dimensions of “state occurrence” and “intensity change”, it

more comprehensively reveals the complex impact of relocation on the psychological stress of left-behind women. Several existing empirical studies on special rural groups in China have adopted this complementary variable measurement method due to similar survey respondent constraints and variable characteristics [62-63].

Explanatory variable

The core explanatory variable in this study is relocation status (RS), a binary variable where participating in the relocation program is coded as 1, and non-participation is coded as 0. This variable serves to identify the net effect of this policy intervention on the psychological stress of left-behind women.

Control variable

Individual characteristics are key factors for measuring a person's capacity to cope with change, while household livelihood capital represents a core resource for resisting risks. Therefore, the study controls for the left-behind women's age (AGE) and education level (EDU), as well as household livelihood capital indicators including the number of laborers (LAB), number of healthy individuals (HEA), per capita operational land area (LAND), per capita fixed assets (FIX), household gift expenditure (GIFT), cooperative membership (COOP), household savings (SAVE), and access to loans (LOAN). The variable for the distance between the

original residence and the township (DIS) controls for the psychological costs associated with geographical isolation. Regional dummy (REG) variables capture unobserved differences across counties and cities in terms of economic development, policy implementation, and cultural practices, ensuring the estimation results are not confounded by regional heterogeneity.

Instrumental variable

Relocation decisions are not random and often involve self-selection issues, leading to estimation bias. To address endogeneity concerns, this study selects the quality of cultivated land in the place of origin as an instrumental variable for relocation behavior. Its rationality lies in three aspects: first, cultivated land quality is a core criterion for the government to identify and select resettled households. The poorer the quality of cultivated land and the harsher the agricultural production conditions in the original residence, the higher the probability that a household will be included in the relocation program, which satisfies the requirement of strong correlation between the instrumental variable and the endogenous variable; second, cultivated land quality is mainly determined by long-term natural conditions such as geology, terrain, and climate, and is a predetermined variable. It is determined before households make relocation decisions and is not adversely affected by the current psychological state of the household. This study has fully incorporated the current household's financial capital, physical capital, and social capital

into the control variables, thereby effectively blocking the impact of cultivated land quality on psychology through the pathway of “current economic level”. Third, under the background of the implementation of China’s resettlement policy, cultivated land quality is an important indicator of eligibility for relocation. Poorer-quality cultivated land always corresponds to a higher probability of relocation, and cultivated land quality is very difficult to improve. There will be no situation where households that originally intended to relocate decide not to participate due to improved cultivated land quality, which satisfies the monotonicity assumption. This study also tests the key assumptions of a valid instrumental variable through methods such as placebo tests, and the results all confirm the validity of the instrumental variable.

Mechanism variables

To unpack the black box of how relocation influences psychological stress, this study introduces changes in consumption expenditure (CON) and perceived livelihood risk (RISK) as mediating variables. This approach aims to empirically test the previously theorized pathways. The selection of these two variables is based on the following rationale. Changes in consumption expenditure directly measure the economic pressure households face after relocation, providing a clear indicator of the “cost shock” pathway. The self-assessed livelihood risk captures the household’s perception of uncertainty regarding future income stability,

reflecting the transformational risks inherent in shifting from agricultural to non-agricultural livelihoods, and thus represents the core of the “risk shock” pathway. By testing the mediating effects of these variables in the relationship between relocation and stress, this approach can accurately reveal the underlying psychological transmission mechanisms and answer the critical question of how relocation leads to increased stress.

Moderator Variable

This study introduces support policies as a moderating variable to explore whether external support can effectively mitigate the negative psychological effects of relocation. Its theoretical basis lies in the fact that stress stems not only from negative life events themselves but also depends on the amount of resources individuals possess when coping with such events, and support policies are precisely such a critical external resource. This variable is comprehensively constructed based on a set of specific questions related to follow-up support in the questionnaire. The questionnaire asked the surveyed households whether they actually received supportive measures led by the government or communities after relocation, such as employment support, economic subsidies, social integration support, and public services. If a surveyed household reported receiving more than two types of the aforementioned support, it was identified as having received substantial “support policies,” and the variable was coded as 1;

otherwise, it indicated no access to any substantive support, and the variable was coded as 0. This variable does not refer to a single policy but represents a comprehensive package of supportive policies following poverty alleviation relocation.

The specific definitions and descriptive statistical analysis of each variable are detailed in table 1. Regarding the core variables, the average score of work stress among left-behind women is 2.75 with a standard deviation of 1.51, indicating that the sample as a whole is at a moderate stress level, but there are significant differences among individuals. The mean value of life stress is 0.46, meaning that nearly half of the respondents feel stressed in daily life. Compared with existing studies on rural women in China, the reported rate of life stress in the sample is at an upper-middle level [62], which may be related to the social transformation brought about by relocation and the special ethnic cultural background. The mean value of relocation behavior is 0.394, which is consistent with our sampling design in key relocation areas, ensuring that both the treatment group and the control group have sufficient sample sizes for analysis. The average score of cultivated land quality, used as an instrumental variable, is only 2.71 with a small standard deviation, which confirms the objective reality of the overall poverty of cultivated land resources in the study area and satisfies the prerequisite that it has a strong correlation with relocation decisions as an instrumental variable. This distribution characteristic is highly consistent with the existing description of

Southwest China as a typical mountainous area with scarce cultivated land resources.

Table 1 Variable descriptions and descriptive statistics

Variable category	Variable	Definition	Mean	Standard Deviation	Max	Min
Dependent variable	WS	Self-rated work stress (1=No stress to 5=Very high stress)	2.75	1.51	5	1
	LS	Self-rated life stress (1=Stressed; 0=Not stressed)	0.460	0.499	1	0
Explanatory variable	RS	Whether household was relocated (1=Yes; 0=No)	0.394	0.489	1	0
Individual characteristics	AGE	Age	3.805	0.329	19	64
	EDU	Education level (1=Illiterate; 0=Non-illiterate)	0.472	0.498	1	0
Human capital	LAB	Number of laborers in the household	2.748	1.722	0	8
	HEA	Number of healthy individuals in the household	2.746	1.890	0	9
Natural capital	LAND	Operational land area / total household members	1.554	1.271	0	6.17
Physical capital	FIX	Total household fixed assets / Total household members	9.128	1.143	142.85	37500
Social capital	GIFT	Total household gift expenditure in 2021	8.115	1.825	0	35000
	COOP	Whether household participates in a cooperative (1=Yes; 0=No)	0.521	0.499	1	0
Financial capital	SAVE	Total current household savings	6.735	4.806	0	200000
	LOAN	Whether the household has access to loans (1=Yes; 0=No)	0.488	0.500	1	0
Distance Feature	DIS	Logarithm of the distance between the original residence and the township	2.768	0.776	1	90
Regional dummy variable	REG	1=Butuo; 2=Meigu; 3=Mianning; 4=Xichang; 5=Xide; 6=Zhaojue	—	—	—	—
Instrumental variable	QUA	Quality of cultivated land (1=Very poor; 2=Poor; 3=Average; 4=Good; 5=Very good)	2.711	0.562	5	1

Mediating variable	CONS	Change in consumption expenditure compared to before (1=Decreased a lot; 2=Decreased somewhat; 3=Similar; 4=Increased somewhat; 5=Increased a lot)	2.614	1.399	5	1
	RISK	Self-rated household risk (1=Very low; 2=Low; 3=Average; 4=High; 5=Very high)	2.393	1.320	5	1
Moderating variable	SUP	Whether received substantial support policies (1=Yes; 0=No)	0.469	0.499	1	0

Model specifications

Baseline regression model

To examine the impact of relocation on the psychological stress of left-behind women in the poverty alleviation relocation program, the baseline regression model (Oprobit model) is specified as follows:

$$\text{Stress}_{ik} = \beta_0 + \beta_1 \text{Relocation}_{ik} + \gamma Z_{ik} + \delta \text{Region}_k + \varepsilon_{ik} \quad (1)$$

In this equation (1), Stress_{ik} represents the psychological stress of left-behind women in the relocation program, Relocation_{ik} indicates whether relocation has occurred, Z_{ik} denotes the vector of control variables, Region_k denotes the regional dummy variable, and ε_i is the random error term. The coefficients to be estimated are β , θ , γ and δ . Assuming $\varepsilon \sim N(0,1)$, the model can be expressed as:

$$P(\text{Stress}_{ik} = 1|X) = P(\text{Stress}_{ik}^* \leq r_0|X) \quad (2)$$

$$= \Phi(r_0 - \beta_1 \text{Relocation}_{ik} - \gamma Z_{ik}) \quad (3)$$

$$P(\text{Stress}_k = 2|X) = P(r_0 \leq \text{Stress}_{ik}^* \leq r_1|X) \quad (4)$$

$$= \varphi(r_1 - \beta_1 \text{Relocation}_{ik} - \gamma Z_{ik}) - \varphi(r_0 - \beta_1 \text{Relocation}_{ik} - \gamma Z_{ik}) \quad (5)$$

.....

$$P(\text{Stress}_{ik} = 5|X) = P(r_3 \leq \text{Stress}_{ik}^*|X) \quad (6)$$

$$= 1 - \varphi(r_3 - \beta_1 \text{Relocation}_{ik} - \gamma Z_{ik} - \delta \text{Region}_k) \quad (7)$$

In equations (2) - (7), r_0 , r_1 , r_2 , and r_3 are parameters to be estimated, satisfying the condition $r_0 < r_1 < r_2 < r_3$. The model estimates these parameters by constructing a likelihood function for the self-assessed stress level of each surveyed left-behind woman in the relocation program and then applying the maximum likelihood estimation method.

Instrumental variable model

This study constructs the instrumental variable model on the basis of the baseline regression model.:

$$\widehat{\text{Relocation}}_{ik} = \alpha_0 + \alpha_1 IV + \gamma Z_{ik} + \delta \text{Region}_k + \varepsilon_{ik} \quad (8)$$

$$\text{Stress}_{ik} = \beta_0 + \beta_1 \widehat{\text{Relocation}}_{ik} + \gamma Z_{ik} + \delta \text{Region}_k + \varepsilon_{ik} \quad (9)$$

In the first stage, the instrumental variable can isolate the exogenous component of the endogenous variable to generate the fitted value $\widehat{\text{Relocation}}_{ik}$, in the second stage of the model, other coefficients are estimated by substituting the fitted value $\widehat{\text{Relocation}}_{ik}$ into the model. In the formula, $\widehat{\text{Relocation}}_{ik}$ denotes the fitted value, IV represents the instrumental variable,

$\widehat{\text{Relocation}}_{ik}$ indicates resettlement for poverty alleviation, other variables are defined as in equation (1).

Mediation effect model

Following the procedures for testing mediating effects proposed by Wen et al. [64-65], this study constructs the following estimation models:

$$\text{Stress}_{ik} = \beta_0 + \beta_1 \widehat{\text{Relocation}}_{ik} + \beta_2 Z_{ik} + \beta_3 \text{Region}_k + \varepsilon_{ik} \quad (10)$$

$$\text{Mediator}_{ik} = \gamma_0 + \gamma_1 \widehat{\text{Relocation}}_{ik} + \gamma_2 Z_{ik} + \gamma_3 \text{Region}_k + \varepsilon'_{ik} \quad (11)$$

$$\text{Stress}_{ik} = \delta_0 + \delta_1 \widehat{\text{Relocation}}_{ik} + \delta_2 \text{Mediator}_{ik} + \delta_3 Z_{ik} + \delta_4 \text{Region}_k + \varepsilon''_{ik} \quad (12)$$

where Mediator_{ik} represents the mediating variable. All other variables are defined as in equation (1).

Moderation effect model

To test the moderating effect of the follow-up support policies in the relocation program, the test model is specified as follows:

$$\text{Stress}_{ik} = \gamma_0 + \gamma_1 \widehat{\text{Relocation}}_{ik} + \gamma_2 \text{policy}_{ik} + \gamma_3 \widehat{\text{Relocation}}_{ik} * \text{policy}_{ik} + \gamma_4 Z_{ik} + \gamma_5 \text{Region}_k + \varepsilon_{ik} \quad (13)$$

where policy_{ik} represents the follow-up support policies. All other symbols share the same definitions as in equation (1). This study uses the interaction term between “household relocation

behavior” and “follow-up support policies” to test the moderating effect of these policies on the pathway through which household relocation behavior influences the psychological stress of left-behind women.

All regression models in this study adopt robust standard errors to mitigate the heteroscedasticity issue. Considering that resettled households within the same county are exposed to highly similar poverty alleviation relocation policies, supporting conditions in resettlement areas, industrial support levels and community governance environments, the sample observations within the same county may have intra-group correlation. If only ordinary robust standard errors are used, the standard errors will be underestimated and the null hypothesis will be excessively rejected. To control for intra-regional correlation and improve the reliability of statistical inferences, standard errors are clustered at the county level in all empirical analyses of this study. This treatment is consistent with the mainstream econometric norms in the field of poverty alleviation relocation and regional policy evaluation, which can effectively absorb unobservable fixed shocks at the regional level and enhance the accuracy of coefficient significance tests and the robustness of results.

Empirical results and analysis

Baseline regression results

Columns (1) to (3) in table 2 present the stepwise regression results examining the impact of relocation on the work stress of left-behind women. Column (1), which does not control for any variables, shows a coefficient of 2.950 for relocation behavior. This result is highly significant at the 1% level ($p < 0.01$), initially indicating a strong association between relocation and increased work stress. Column (2) introduces regional dummy variables to control for regional heterogeneity based on column (1), and the estimation results remain robust ($p < 0.01$). In column (3), which includes all control variables, the coefficient for relocation decreases to 1.342 but remains significant at the 1% level ($p < 0.01$). These findings demonstrate that individual and household characteristics partially explain the impact of relocation on work stress, yet relocation itself maintains an independent and significant stress-promoting effect.

In the regression analysis for life stress, column (4) shows a coefficient of 1.281 ($p < 0.01$) for relocation behavior. In column (5), which introduces regional dummy variables, the coefficient slightly increases to 1.333 ($p < 0.01$). When all control variables are included in column (6), the coefficient for the impact of

relocation on the life stress of left-behind women further attenuates to 0.613, yet remains significant at the 1% level ($p < 0.01$). This indicates that although the control variables absorb part of the effect, the positive impact of relocation on life stress remains robust.

The study also finds that the accumulation of human capital, financial capital, and social capital plays an important role in alleviating the psychological stress of left-behind women. Good health status, sound household assets and savings levels, and smooth access to financing all help reduce their work and life stress. Furthermore, support from social networks can also exert a positive influence, although it is important to note that some forms of social participation may increase psychological stress by imposing additional responsibilities.

Table 2 Baseline regression results of the impact of relocation on the mental health of left-behind women

	(1) WS	(2) WS	(3) WS	(4) LS	(5) LS	(6) LS
RS	2.950*** (0.164)	3.001*** (0.174)	1.342*** (0.195)	1.281*** (0.099)	1.333*** (0.101)	0.613*** (0.167)
AGE			-0.125 (0.153)			0.056 (0.155)
EDU			-0.263 (0.177)			0.028 (0.212)
LAB			-0.037 (0.063)			-0.108 (0.088)
HEA			-0.271*** (0.060)			-0.014 (0.085)
LAND			-0.090			-0.012

			(0.079)			(0.103)
FIX			-0.385***			-0.288***
			(0.074)			(0.097)
GIFT			-0.046			-0.181*
			(0.087)			(0.098)
COOP			0.268*			-0.137
			(0.155)			(0.260)
SAVE			-0.068***			-0.035**
			(0.015)			(0.018)
LOAN			-0.796***			-0.890***
			(0.204)			(0.337)
DIS			0.084			-0.009
			(0.096)			(0.113)
REG	NO	YES	YES	NO	YES	YES
Pseudo R ²	0.298	0.304	0.485	0.165	0.181	0.425
Log	-858.06	-851.67		-448.76	-440.28	-405.716
pseudolikelihood	2	7	-629.535	4	8	
N	779	779	779	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; LS = Self-rated life stress; AGE = Age; EDU = Education level; LAB = Number of laborers in the household; HEA = Number of healthy individuals in the household; LAND = Operational land area per household member; FIX = Total household fixed assets per household member; GIFT = Total household gift expenditure in 2021; COOP = Whether household participates in a cooperative; SAVE = Total current household savings; LOAN = Whether the household has access to loans; DIS = Logarithm of the distance between the original residence and the township; REG = Regional dummy variable.

The baseline regression reveals the overall impact of relocation on the psychological stress of left-behind women, yet average effects may mask heterogeneous responses across different groups. To uncover group differences in relocation effects, this study further conducts heterogeneity analysis based on household income and industrial development in resettlement areas. Results in table 3 show distinct psychological challenges faced by different groups during relocation adaptation.

In the high-income group, relocation demonstrates a strong positive effect on work stress ($\beta = 5.476$, $p < 0.01$) while showing a significant negative effect on life stress ($\beta = -4.282$, $p < 0.01$). This finding suggests that while high-income households can leverage resource advantages to alleviate daily life pressures, they face higher expectations and competitive pressures in the work domain. In stark contrast, relocation exerts significant positive effects on both work stress ($\beta = 1.480$, $p < 0.01$) and life stress ($\beta = 1.747$, $p < 0.01$) in the low-income group, highlighting the comprehensive vulnerability of this group under resource constraints.

Subgroup regression based on industrial layout and development near resettlement areas reveals deeper-level differences. For households with access to local industrial support, relocation shows a relatively moderate impact on work stress ($\beta = 1.067$, $p < 0.01$) and significantly alleviates life stress ($\beta = -1.201$, $p < 0.05$). This indicates that an industrial foundation can provide flexible employment opportunities for relocated families, effectively reducing psychological pressure. In comparison, households lacking industrial support face more severe challenges, with a considerably higher coefficient for work stress ($\beta = 3.412$, $p < 0.01$) and non-significant alleviation of life stress ($\beta = 0.204$), reflecting greater psychological pressure during their livelihood transition.

Table 3 Regression results for heterogeneity

WS	LS	WS	LS
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	High_ incom e	Low_ i ncom e	High_ i ncom e	Low_ i ncom e	With_ indus try	No_in dustr y	With_i ndustr y	No_in dustr y
RS	5.476* **	1.480* **	-4.282* **	1.747** *	1.067* **	3.412* **	-1.201* *	0.204
	(0.331)	(0.201)	(0.289)	(0.189)	(0.179)	(0.725)	(0.409)	(0.377)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
REG	YES	YES	YES	YES	YES	YES	YES	YES
Pseudo R ²	0.321	0.527	0.756	0.698	0.402	0.622	0.705	0.707
Logpseudolikeli hood	-160.2 7	-398.3 7	-179.4 4	-216.9 8	-395.7 7	-195.5 3	-76.12	-71.76
N	359	420	359	420	414	365	414	365

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; LS = Self-rated life stress; REG = Regional dummy variable.

Endogeneity test

Instrumental variable checks

To address the potential endogeneity of relocation behavior, this study employs the instrumental variable (IV) method with cultivated land quality as the instrumental variable to resolve endogeneity issues in the model. The results of the CMP model are reported in table 4. The first-stage regression reveals a significant negative relationship between cultivated land quality and participation in poverty alleviation relocation ($\beta = -0.104$, $p < 0.01$), indicating that villages and households with poorer cultivated land quality are more likely to be included in the relocation program. The F-statistic of the first-stage model is 28.63 and the Cragg-Donald Wald F-statistic is 39.17, both of which are greater than the critical value at the 10% level, demonstrating the absence of a

significant weak instrumental variable problem. Further endogeneity tests (insig_2 and atanhrho_12) are all significant at the 1% or 5% level, which confirms the existence of endogeneity in relocation behavior and the necessity of using the instrumental variable method.

The second-stage regression results indicate that, after controlling for endogeneity, the positive effect of relocation on work stress increases to 3.778 ($p < 0.01$), and its positive effect on life stress is 2.211 ($p < 0.01$). This test further verifies that the stress-inducing effect of relocation on left-behind women remains significant and robust. Although the baseline regression controls for a rich set of observable variables, numerous unobservable household characteristics simultaneously influence relocation decisions and psychological stress, leading to the underestimation of the negative effects of relocation. This study further employs the 2SLS model for robustness checks. The F-statistic of the first-stage model is 23.11 and the Cragg-Donald Wald F-statistic is 33.12, both of which are greater than the critical value at the 10% level, once again demonstrating the absence of a significant weak instrumental variable problem. The results (table 5) still support the original hypothesis. After accounting for the potential endogeneity of relocation decisions, the correlation between relocation and stress may be underestimated, which indicates that the instrumental variable affects stress only through its impact on relocation decisions and is unrelated to unobservable psychological

traits or implicit support. It can effectively isolate self-selection bias and correct the model estimation bias to a certain extent. However, it should be noted that instrumental variable estimation relies on specific identification assumptions, and its results should be interpreted as complementary evidence for the baseline regression rather than the sole characterization of the causal effect.

Table 4 Endogeneity Tests of Relocation's Impact on the Mental Health of Left-behind Women

	First stage RS	Second WS	First stage RS	Second LS
RS		3.778*** (0.11)		2.211*** (0.59)
QUA	-0.104*** (0.01)		-0.104*** (0.0)	
Controls	YES	YES	YES	YES
REG	YES	YES	YES	YES
lnsig_2		-1.301*** (0.0)		-1.301*** (0.0)
atanrho_12		-1.491*** (0.1)		-0.517** (0.23)
N	779	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; LS = Self-rated life stress; QUA=Quality of cultivated land; REG = Regional dummy variable.

Table 5 Robustness Test of Endogeneity in the Impact of Relocation on the Mental Health of Left-behind Women

	First stage RS	Second WS	First stage RS	Second LS
RS		4.380*** (0.82)		0.846** (0.29)
QUA	-0.104*** (0.02)		-0.104*** (0.0)	
Controls	YES	YES	YES	YES
REG	YES	YES	YES	YES
cons	1.195*** (0.20)	0.224*** (1.48)	1.195*** (0.20)	0.478* (0.523)
N	779	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the

acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; LS = Self-rated life stress; QUA=Quality of cultivated land; REG = Regional dummy variable.

Robustness test

To ensure the reliability of the findings, this study conducts robustness checks using multiple methods, with results presented in table 6. First, the analysis replaces the measurement method of the dependent variable, using “sense of happiness” (HAP) and “insomnia status” (INS) as alternative measures. Second, considering the potential influence of regional heterogeneity on the estimation results, the study excludes samples from Xichang City, which exhibits significant differences in geographical location and economic development level. Finally, to eliminate potential bias from specific subgroups, households primarily reliant on government subsidies are removed from the sample. The results from these three different robustness checks consistently demonstrate that relocation has a significant negative impact on the psychological stress of left-behind women, fully confirming the reliability and robustness of the core findings.

Table 6 Robustness tests of relocation’s impact on left-behind women’s mental health

Variable	Change variable		Exclude zone		Exclude samples	
	HAP	INS	HAP	INS	HAP	INS

		1.434**		0.525**	0.473**	0.556**
RS	-1.100*	*	1.327**		*	
	(0.187)	(0.186)	(0.193)	(0.171)	(0.238)	(0.175)
Controls	YES	YES	YES	YES	YES	YES
REG	YES	YES	YES	YES	YES	YES
R ²	0.689	0.555	0.477	0.247	0.473	0.183
Log likelihood	-244.6	-540.77	-573.33	-361.66	-500.09	-348.21
N	69	7	1	8	9	9
	779	779	694	694	664	664

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; HAP = sense of happiness; INS = insomnia status; REG = Regional dummy variable.

Mechanism analysis

Table 7 presents the results of the mediating mechanism test examining how relocation affects the work stress of left-behind women through household consumption level and household livelihood risk level. The results in Column (1) reconfirm that the total effect of relocation on work stress is significantly positive ($\beta = 1.342$, $p < 0.01$). The results in Column (2) show that relocation has a significantly positive impact on the household consumption level of left-behind women ($\beta = 1.349$, $p < 0.01$). After including both relocation and consumption level variables in Column (3), consumption level itself exerts a significantly positive effect on work stress ($\beta = 0.836$, $p < 0.01$), while the coefficient of the direct effect of relocation decreases from 1.342 to 0.788 ($p < 0.01$). This indicates that consumption level plays a role of partial mediator

between relocation and work stress, meaning that relocation increases work stress partly by raising household consumption level.

Table 7 The Impact of Poverty Alleviation Relocation on the Work Stress of Left-behind Women: A Path Analysis Based on Consumption Level

	(1) WS	(2) CONS	(3) WS
RS	1.342*** (0.195)	1.349*** (0.178)	0.788*** (0.213)
CONS			0.836*** (0.105)
Controls	YES	YES	YES
REG	YES	YES	YES
Pseudo R ²	0.485	0.453	0.539
N	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; CONS=Change in consumption expenditure compared to before; REG = Regional dummy variable.

Similarly, from the perspective of the household livelihood risk level path (table 8), Column (2) shows that relocation significantly increases the livelihood risks faced by households ($\beta = 0.609$, $p < 0.01$). Column (3) reveals that household livelihood risk level has a significantly positive predictive effect on work stress ($\beta = 0.315$, $p < 0.01$), and at the same time, the coefficient of the direct effect of relocation decreases from 1.342 to 1.216 ($p < 0.01$). This result verifies that household livelihood risk level constitutes another significant mediating channel through which relocation affects

work stress. Based on the results in table 7 and table 8, relocation not only directly exacerbates work stress but also exerts a positive effect on work stress through two indirect paths: raising household consumption level and increasing household livelihood risk level.

Table 8 The Impact of Poverty Alleviation Relocation on the Work Stress of Left-behind Women: A Path Analysis Based on Household Livelihood Risk Level

	(1) WS	(2) RISK	(3) WS
RS	1.342*** (0.195)	0.609*** (0.144)	1.216*** (0.199)
RISK			0.315*** (0.052)
Controls	YES	YES	YES
REG	YES	YES	YES
Pseudo R ²	0.485	0.227	0.503
N	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; RISK=Self-rated household risk; REG = Regional dummy variable.

Table 9 presents the results of the mediating mechanism analysis with life stress as the dependent variable. In Column (1), the total effect of relocation on life stress is also significantly positive ($\beta = 0.613$, $p < 0.01$). For the consumption level path, Column (2) confirms that poverty alleviation relocation exerts a significant positive effect on consumption level ($\beta = 1.349$, $p < 0.01$). In Column (3), the coefficient of consumption level on life stress is 0.275 ($p < 0.01$), and the coefficient of the direct effect of

relocation decreases from 0.613 to 0.395 ($p < 0.05$). This confirms that consumption level has a significant partial mediating effect in the relationship between relocation and life stress.

Table 9 The Impact of Poverty Alleviation Relocation on the Life Stress of Left-behind Women: A Path Analysis Based on Consumption Level

	(1) LS	(2) CONS	(3) LS
RS	0.613*** (0.167)	1.349*** (0.178)	0.395** (0.179)
CONS			0.275*** (0.085)
Controls	YES	YES	YES
REG	YES	YES	YES
Pseudo R ²	0.425	0.453	0.526
N	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; LS = Self-rated life stress; CONS=Change in consumption expenditure compared to before; REG = Regional dummy variable.

For the household livelihood risk level path (table 10), Column (2) shows that household livelihood risk level has a significantly positive impact on life stress ($\beta = 0.222$, $p < 0.01$). After including this mediating variable, the coefficient of the direct effect of relocation on life stress decreases from 0.613 to 0.484 ($p < 0.01$). The results of table 9 and table 10 generally indicate that for life stress, both household consumption level and household livelihood risk level constitute significant mediating paths through which relocation exerts its impact. Their mechanism of action is

consistent with that on work stress in terms of direction, yet there are differences in the magnitude of the coefficients.

Table 10 The Impact of Poverty Alleviation Relocation on the Life Stress of Left-behind Women: A Path Analysis Based on Household Livelihood Risk Level

	(1) LS	(2) RISK	(3) LS
RS	0.613*** (0.167)	0.609*** (0.144)	0.484*** (0.172)
RISK			0.222*** (0.054)
Controls	YES	YES	YES
REG	YES	YES	YES
Pseudo R ²	0.425	0.427	0.561
N	779	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; LS = Self-rated life stress; RISK=Self-rated household risk; REG = Regional dummy variable.

Empirically, prior to poverty alleviation relocation, households had long adhered to a semi-subsistence livelihood model, with basic needs such as food, fuel, and housing highly reliant on self-production and self-consumption, resulting in extremely low pressure from household cash outlays. Following centralized resettlement, payments for water, electricity, gas, and daily consumer goods have fully shifted to market-based and monetized modes, leading to a sharp surge in rigid expenses and the emergence of a passive consumption shock. As the primary managers of household daily income and expenditure and core

bearers of care responsibilities, left-behind women are more sensitive to the abrupt changes in consumption structure and the widening cash shortfalls. The rising consumption costs directly translate into persistent economic anxiety and life stress. Additionally, poverty alleviation relocation has physically severed farmers' connection with traditional means of production such as cultivated land and forest land. Non-agricultural employment generally requires basic skills and stable working hours. These requirements are highly mismatched with the characteristics of left-behind women, who often have low educational attainment, strong constraints from family caregiving, and limited access to vocational training. This mismatch fosters a strong perception of livelihood uncertainty. Livelihood risks directly drive work stress and, at the same time, transmit to life stress through the deterioration of household expectations for economic security. This is consistent with the conclusions of scholars' research that consumption burdens and perceptions of livelihood risk are important transmission paths for mental health [66], and it also addresses the deficiency of insufficient gender perspective in existing literature.

Moderation effect analysis

Table 11 examines the moderating effect of policy support on the relationship between relocation and the psychological stress of left-behind women. The regression results show that the interaction term (INTE) between relocation behavior and policy support has a significant negative moderating effect on both work stress ($\beta = -0.288$, $p < 0.01$) and life stress ($\beta = -0.217$, $p < 0.05$). This indicates that stronger policy support can effectively mitigate the negative impact of relocation on the psychological well-being of left-behind women. Specifically, among the group receiving higher levels of policy support, the increase in psychological stress induced by relocation is significantly smaller. The existence of the moderating effect indicates that external supportive policies constitute an important boundary condition for mitigating the potential negative psychological consequences of relocation. As an external protective resource, support policies can supplement lost resources, stabilize livelihood expectations, and reduce psychological stress among left-behind women.

Table 11 Results of the test for the moderating effects of support policies

	(1) WS	(2) LS
RS	2.165** (0.395)	1.178*** (0.374)
SUP	-0.157*** (0.056)	-0.107 (0.077)
INTE	-0.288***	-0.217**

	(0.091)	(0.104)
Controls	YES	YES
REG	YES	YES
Pseudo R ²	0.506	0.264
Log pseudolikelihood	-603.408	-395.647
N	779	779

Note: 1. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. The values in parentheses represent robust standard errors. 2. The full meanings of the acronyms are as follows: RS = Whether household was relocated; WS = Self-rated work stress; LS = Self-rated life stress; SUP = Whether received substantial support policies ; INTE=interaction term; REG = Regional dummy variable.

Conclusions and policy implications

Conclusions

The core findings of this study indicate that relocation exerts a significant inhibitory effect on the mental health of left-behind women. This conclusion reveals the complex socio-psychological adaptation process underlying large-scale population migration. Relocation not only alters living spaces but also triggers a systematic restructuring of individual livelihood modes, social networks, family roles, and psychological identity.

Relocation entails a fundamental transformation of the household livelihood foundation and amplifies the transmission effect of stress. Our findings also form a meaningful dialogue with the broader migration research. The conclusion that poverty alleviation relocation exacerbates psychological stress is consistent with numerous studies on involuntary or passive migration, which

have generally found that the loss of control and the breakdown of social networks during the migration process lead to psychological costs [67-69]. The shift from land-dependent farming livelihoods to fully market-based non-farm livelihoods renders the core livelihood capital of agricultural production skills and knowledge rapidly devalued in an urban context, creating a skills mismatch dilemma. Concurrently, rigid increases in the cost of living transition household finances from a relatively self-sufficient and resilient state to a cash-flow-dependent vulnerable state. Furthermore, relocation does not change the essential reality of women being left behind; instead, it intensifies their family responsibilities immensely. Without adequate support, this leads to severe role overload for left-behind women, consequently triggering psychological stress.

Heterogeneity analysis reveals that poverty alleviation relocation exerts significantly heterogeneous impacts on the life stress of households with different income levels. For high-income households, relocation typically means moving into communities with more abundant resources; the improvement of infrastructure such as transportation, medical care and education enables them to access needed resources more conveniently in daily life, reducing time and economic pressures. For low-income households,

relocation leads to a rigid rise in living costs, especially the increase in expenses such as housing, water, electricity and transportation, resulting in a marked surge in economic pressure. Although relocation may provide new employment opportunities for these households, their lack of sufficient wealth reserves makes them highly sensitive to daily expenses, and the rise in living costs often inflicts greater psychological burdens on them. In addition, the significant disparities associated with the presence or absence of industrial support in resettlement areas provide direct empirical evidence for the spatial mismatch theory, and underscore the fundamental importance of addressing the geography-employment matching issue for promoting the psychosocial integration of migrants.

Poverty alleviation relocation exacerbates the psychological stress of left-behind women by increasing household consumption costs and raising perceptions of livelihood risk. This finding not only uncovers the specific mechanisms through which poverty alleviation relocation affects mental health, but also confirms the core role of economic pressure transmission and uncertainty anxiety in the psychological adaptation process to major life events [70-71]. The consumption cost path highlights the shock of financial rigidification brought about by the transition from

subsistence agriculture to a cash economy, while the risk perception path reflects the loss of future sense of security stemming from the shift from stable farming to volatile non-agricultural employment. These have become the key sources of psychological stress for left-behind women in the context of relocation, and provide a clear mechanistic explanation for understanding the complex psychosocial effects of relocation policies.

The effects of poverty alleviation relocation on the psychological stress of left-behind women are multidimensional, and its impact paths are intertwined with factors such as consumption level and livelihood risk perception. Therefore, when formulating supportive policies for poverty alleviation relocation, special consideration should be given to inter-group heterogeneity, particularly the pressures borne in the face of rising living costs and mismatched employment opportunities. The effective moderating effect of supportive policies validates the classic theory of institutional social support as a stress buffer [72], and provides positive empirical evidence for the Chinese characteristic policy system of follow-up support. This is consistent in direction with the conclusions of studies in other countries that integration policies can improve migrants' mental health.

Policy implications

Based on the pathway analysis of how relocation affects the psychological stress of left-behind women, this study proposes targeted policy recommendations focusing on mitigating cost shocks and enhancing risk resilience. These measures aim to support relocated households in achieving a stable livelihood transition, thereby effectively alleviating psychological stress.

First, construct a comprehensive support system that integrates employment support and psychological services. Integrate resources to establish an employment service platform and provide skills training aligned with local industries; at the same time, incorporate mental health services into community public services, and build a support network by setting up consulting points and training grassroots personnel.

Second, implement differentiated support strategies. Strengthen living security and subsidies for low-income households; focus on employment guidance and financial support for middle-income households; optimize public services and channels for social participation for high-income households to alleviate psychological gaps.

Third, vigorously develop local industries and inclusive

childcare services, and optimize the living and development environment for left-behind women. In response to the data showing that left-behind women face few employment opportunities and heavy caregiving burdens, it is recommended to develop characteristic industries around resettlement areas to promote nearby employment; simultaneously accelerate the construction of inclusive childcare and elderly care services to directly reduce family caregiving pressure and improve employment feasibility.

Fourth, innovate community governance to reduce living costs. Encourage resettlement communities to explore innovative models for reducing residents' living costs, and enhance community cohesion and reduce household daily living costs through community cooperatives, public canteens, or centralized purchasing, so as to effectively alleviate household economic pressure.

Research Limitations and Future Prospects

Despite conducting a rigorous empirical analysis that identifies the inhibitory effect of poverty alleviation relocation on the mental health of left-behind women and uncovers the mediating transmission paths of consumption costs and livelihood risks, this

study provides empirical support for the formulation of follow-up support policies for relocation. Nevertheless, this study still has several limitations worthy of in-depth discussion due to constraints on data availability, research methods and practical survey conditions. Future research can be improved in the following directions:

First, limitations of data and samples. The cross-sectional data adopted in this study cannot capture the dynamic evolution process of their psychological adaptation. Although the survey made every effort to reduce potential impacts such as self-report and recall biases through anonymous commitments, neutral questioning, and standardized training of investigators, the implicit effects of such biases cannot be completely eliminated. Future research can introduce a multi-wave panel survey design to systematically depict the dynamic changes in the mental health of left-behind women. It can also optimize the survey design by conducting cross-cultural and cross-regional comparative studies, and incorporate objective indicators to assist in verifying the reliability of subjective data and reduce measurement errors.

Second, limitations of endogeneity treatment. To mitigate the endogeneity of self-selection in relocation behavior, this study uses the quality of cultivated land in the place of origin as the

instrumental variable, but there are still two shortcomings: first, the measurement of the instrumental variable is based on respondents' evaluations, which may have certain deviations compared with the objective physical and chemical indicators of land; second, it is difficult for any observational study to completely eliminate the impact of endogeneity. Although the model has controlled for multi-dimensional covariates, it still cannot completely avoid omitted variable bias. In addition, future research can combine remote sensing technology data and objective soil fertility grade data registered by agricultural and rural departments to construct more accurate instrumental variables and further mitigate endogeneity bias.

Third, limitations of mechanism analysis. The core of the mechanism analysis in this study is to reveal the transmission paths of consumption costs and livelihood risks between relocation and the psychological stress of left-behind women, and the causal mediation model based on instrumental variables is not adopted. The reason is that it is extremely difficult to find valid instrumental variables for potentially endogenous mediating variables such as consumption costs and livelihood risks, which to a certain extent limits the causal rigor of mechanism inference. If future research can draw on natural experiments in policy implementation,

regression discontinuity designs or multi-period instrumental variables, or construct exogenous shocks for mediating variables, it will be possible to conduct more rigorous causal identification of the impact paths of relocation.

Abbreviations

COOP	Whether household participates in a cooperative
CONS	Change in consumption expenditure compared to before
DIS	Logarithm of the distance between the original residence and the township
EDU	Education level
FIX	Total household fixed assets per household member
GIFT	Total household gift expenditure in 2021
HAP	Sense of happiness
HEA	Number of healthy individuals in the household
INS	Insomnia status
INTE	Interaction term (relocation $\times$ support policy)
LAB	Number of laborers in the household
LAND	Operational land area per household member
LOAN	Whether the household has access to loans
LS	Self-rated life stress
QUA	Quality of cultivated land
REG	Regional dummy variable
RISK	Self-rated household risk
RS	Whether household was relocated (Relocation Status)
SAVE	Total current household savings
SUP	Whether received substantial support policies
WS	Self-rated work stress

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Authors' contributions

Conceptualization, C.L.; methodology, C.L.; software, C.L.; validation, H.C. and R.Z.; formal analysis, R.Z.; investigation, H.C.; resources, H.C.; data curation, R.Z.; writing-original draft preparation, C.L., H.C. and R.Z.; writing-review and editing, Y.L.; visualization, Y.L.; supervision, Y.L.; project administration, Y.L.; funding acquisition, C.L. All authors have read and agreed to the published version of the manuscript.

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Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations**Ethics approval and consent to participate**

This study was conducted in accordance with the Declaration of Helsinki. Ethics approval for this study was obtained on 10 June

2024, before the field survey was carried out. The study was reviewed and approved by the Academic Ethics Review Subcommittee of the School of Resources and Environment, Xichang University, China. Written informed consent was obtained from participants prior to the interviews. For participants who were unable to sign due to educational level or other reasons, verbal consent was obtained in the presence of a witness, and the witness signed the consent record for confirmation. Participation was voluntary. In addition, privacy and data protection measures were implemented throughout the research process, including anonymization of personal identifiers and secure storage of both paper and electronic data before analysis.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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